

Def. A differentiable function $f: \mathbb{R}^n \rightarrow \mathbb{R}^m$ is called a mapping.

Def. Let $F: \mathbb{R}^n \rightarrow \mathbb{R}^m$ be a mapping and V_p be a tangent vector.

Then, for a curve $\beta: (-\epsilon, \epsilon) \rightarrow \mathbb{R}^m: t \mapsto F(p + t \cdot V)$, define

$$F_*(V_p) \stackrel{\text{def}}{=} \beta'(0)_{F(p)}.$$

The map $F_*: \bigcup_{p \in \mathbb{R}^3} T_p \mathbb{R}^3 \rightarrow \bigcup_{p \in \mathbb{R}^3} T_{F(p)} \mathbb{R}^3$ is called the tangent map of F .

Lemma. Let $F = (f_1, \dots, f_m): \mathbb{R}^n \rightarrow \mathbb{R}^m$ be a mapping. Given a tangent vector V_p ,
Proof.

$$\begin{aligned} F_*(V_p) &= (V_p[f_1], V_p[f_2], \dots, V_p[f_m]) \\ &= (Df_1(p) \cdot V, Df_2(p) \cdot V, \dots, Df_m(p) \cdot V) \end{aligned}$$

$$= DF(p) \cdot V$$

where $DF(p)$ is the Jacobian, $DF(p) = \left(\frac{\partial f_i}{\partial x_j}(p) \right)_{i,j}$.

Note: For fixed $p \in \mathbb{R}^3$, F_{*p} is linear with respect to the vector V at p .

Def. A mapping $F: \mathbb{R}^n \rightarrow \mathbb{R}^m$ is called regular if: for each $p \in \mathbb{R}^3$, F_{*p} is injective.